Nuvaxovid (XFG): Pharmacist Rundown and Counselling Sheet

Recombinant protein, Matrix-M-adjuvanted COVID-19 vaccine; the non-mRNA option in Alberta's 2026–27 program. Sources hyperlinked at each mention. Section 6 explains how to read the efficacy numbers and compares them with the mRNA vaccines. Prepared 9 November 2026.

At a glance

• **Product:** Nuvaxovid (Omicron XFG), Sanofi Vaccines Canada (market-authorization holder since April 2026, taking over from Novavax; Sanofi, 9 Apr 2026). Authorized in Canada 16 Sept 2026; in Alberta program from 1 Oct 2026 (AIP biological page 07.229).

• **Dose:** 0.5 mL IM = 5 µg recombinant spike protein + 50 µg Matrix-M adjuvant. Non-live. No preservative, no egg, latex, blood or animal products.

• **Age:** 12 years and older.

• **Presentations:** single-dose prefilled syringe (no needle) and 5-dose vial. Fridge 2–8 °C, protect from light, do not freeze, do not shake (swirl vials). Punctured vial: 12 h at 2–8 °C or 6 h at ≤25 °C (Product monograph, June 2026).

• **Alberta schedule (healthy, 12+):** one dose, ≥3 months after any previous COVID-19 dose, regardless of how many doses before, including none. **Moderately/severely immunocompromised** with <3 prior doses: 3-dose series (day 0, ≥28 days, then 8 weeks); with ≥3 prior doses: 1 dose ≥3 months after the last.

• **Funded for:** the same groups as mRNA vaccine (≥65; continuing-care residents 12+; 12+ with underlying or immunocompromising conditions, pregnancy or houselessness; First Nations, Métis, Inuit 12+; health-care workers; teachers). Others pay at community pharmacies.

• **Position:** NACI and Alberta express **no preference** for mRNA over protein-subunit vaccine for people 12+ (NACI 2024; NACI fall 2026; AIP 07.229).

1. Why it exists and who asks for it

Nuvaxovid fills three gaps in an mRNA-dominated program: people who had a contraindication or a serious reaction to an mRNA vaccine (including polyethylene glycol allergy, which Nuvaxovid does not contain), people who refuse mRNA on principle but will accept a conventional protein vaccine, and people who found mRNA reactogenicity intolerable. Every pharmacy sees all three. The platform (recombinant protein on an adjuvant) is the same technology family as Shingrix, Arexvy and the hepatitis B and HPV vaccines, which is a useful anchor when patients ask whether it is "new".

2. How it is built

2.1 The antigen

The full-length SARS-CoV-2 spike gene (now from the Omicron XFG lineage) is inserted into a baculovirus, which infects *Spodoptera frugiperda* (Sf9) moth cells; the cells secrete spike trimers that are purified and assembled with polysorbate 80 into ~15–20 nm nanoparticles, each carrying several trimers (Keech et al., NEJM 2020; Product monograph). The spike is locked in its **prefusion** shape by two proline substitutions and by mutating the furin cleavage site, the same structure-based trick used in the RSV vaccines, so the antibodies raised recognize the virus before it fuses with a cell. Unlike mRNA vaccines, which deliver instructions for the patient's own cells to make spike, Nuvaxovid delivers the finished protein; nothing is made in the body and there is no nucleic acid in the product.

2.2 Matrix-M, the adjuvant

Matrix-M is 50 µg of saponins purified from the bark of the Chilean soap-bark tree *Quillaja saponaria*, formulated with cholesterol and phospholipid into 40 nm cage-like particles (two fractions, Matrix-A and

Matrix-C, blended to balance potency and tolerability) (Stertman et al., Hum Vaccin Immunother 2023). It recruits neutrophils, monocytes and dendritic cells to the injection site and draining lymph node within hours, activates antigen-presenting cells, and drives a balanced Th1/Th2 CD4 response with strong T-follicular-helper and germinal-centre activity. In phase 1 the adjuvant was dose-sparing: 5 µg of adjuvanted spike out-performed 25 µg unadjuvanted (Keech 2020). It is in the same saponin family as the QS-21 in AS01 (Shingrix, Arexvy), which explains the similar "adjuvant day" of fatigue and myalgia some patients notice, although Nuvaxovid is less reactogenic than Shingrix.

3. How well it works

3.1 Randomized trials (original strain)

- **PREVENT-19**, 29,949 adults in the US and Mexico (Dunkle et al., NEJM 2022): efficacy 90.4% (95% CI 82.9–94.6) against symptomatic COVID-19 and 100% against moderate-to-severe disease; 92.6% against variants of concern (Alpha-predominant).
- **UK phase 3**, 15,187 adults (Heath et al., NEJM 2021): 89.7% (80.2–94.6) overall; 86.3% against Alpha; 96.4% against the ancestral strain.
- **Adolescents 12–17**, 2,247 participants (Áñez et al., JAMA Netw Open 2023): 79.5% (46.8–92.1) during Delta predominance, with neutralizing titres 1.5-fold higher than in young adults.
- **As a heterologous booster** (COV-BOOST, Munro et al., Lancet 2021): a Nuvaxovid third dose after two doses of AstraZeneca or Pfizer produced robust antibody and T-cell responses, supporting mix-and-match use.

3.2 Real-world effectiveness

- Italy, 20,903 adults receiving a Nuvaxovid primary series during Omicron (Mateo-Urdiales et al., JAMA Netw Open 2023): 50% (40–58) against symptomatic COVID-19 and 31% against any infection; hospitalizations too few to estimate; protection against symptomatic disease was stable to 4 months.
- South Korea, propensity-matched national cohort, Nuvaxovid versus Pfizer as third or fourth dose (Gwak et al., medRxiv 2024, preprint): hazard ratios for severe disease (ICU or death) 0.73 (0.53–1.00) after dose 3 and 1.21 (1.03–1.42) after dose 4, i.e. no consistent difference from mRNA.
- **Updated-strain versions (XBB.1.5 → JN.1 → XFG)** are authorized on immunogenicity, as every current COVID-19 vaccine is; there are no efficacy trials of the XFG formulation for any manufacturer (Product monograph). Expect the same order of protection as the mRNA products: roughly 50% against hospitalization in the first two months, falling to about a third by five months (Wiegand et al., JAMA Intern Med 2026; JAMA Intern Med 2026, Nebraska cohort).

Plain summary: the original Nuvaxovid matched the mRNA vaccines in trials (≈90%), performs comparably as a booster in observational data, and the yearly strain updates are approved on the same antibody-bridging basis as Spikevax and Comirnaty. NACI's 2024 rationale for dropping the mRNA preference was exactly this equivalence plus the larger safety record (NACI 2024).

4. Safety and reactogenicity

4.1 Expected reactions

XBB.1.5 booster data in adults (Product monograph): injection-site pain/tenderness 56%, fatigue 33%, myalgia 29%, headache 22%; fever uncommon (≈6% after dose 2 of the original series). Adolescents report more (pain 77%, fatigue 63%, headache 63%, fever 17%). Reactions begin within a day and resolve in 1–2 days.

4.2 Versus mRNA

- Prospective US/Canada cohort of 1,130 working adults, inverse-probability weighted (Rousculp et al., Vaccines 2024): mean number of systemic reactions 1.8 vs 3.2; three or more systemic reactions 33.9% vs 61.4% for Nuvaxovid vs mRNA. A companion study of 1,367 adults found less local and systemic reactogenicity but no significant difference in ≥50% work impairment (Rousculp et al.,

Vaccines 2024 (burden)). Note: industry-funded.

- COMPARE, a phase 4 double-blind RCT of 1,000 US adults against Moderna's mNEXSPIKE, presented at ESCMID 2026 (conference abstract, not yet peer-reviewed; report): any systemic reaction 83.6% vs 91.6%; severe systemic reactions under 1 in 10 vs about 1 in 5; severe injection-site reactions >75% less frequent.

4.3 Myocarditis and pericarditis

Both are listed as rare in the monograph, with causality not established (Product monograph). The largest clinically reviewed dataset, 102,946 doses in two Australian states (Clothier et al., Eurosurveillance 2024), found 2 myocarditis (1.9 per 100,000 doses) and 20 pericarditis (19.4 per 100,000) reports, no Guillain-Barré, no TTS and no vaccine-attributable deaths. For comparison, mRNA myocarditis in young males after dose 2 was reported at roughly 10–20 per 100,000. NACI expects the risk to be lower still with the one-dose schedule (NACI 2024). Alberta's precaution rules (AIP 07.229): myocarditis within 6 weeks of any COVID-19 vaccine → generally defer further doses (may offer after risk–benefit discussion); pericarditis with normal work-up → may re-immunize once symptom-free and ≥90 days have passed; myocarditis unrelated to vaccination → no deferral. Counsel every recipient to seek care for chest pain, dyspnea or palpitations.

4.4 Contraindications, precautions, special populations

- Contraindicated only for anaphylaxis to a previous dose or to a component; polysorbate 80 is the component most often implicated in hypersensitivity. A prior mRNA reaction is not a contraindication; it is a common reason to choose this product.
- Pregnancy and breastfeeding: may be used in any trimester; NACI notes more safety data exist for mRNA in pregnancy, so this is a shared decision (AIP 07.229).
- Bleeding disorders/anticoagulants: same IM precautions as any vaccine. Defer during acute febrile illness. Vaccinate after COVID-19 infection once recovered and non-infectious (after MIS-A/MIS-C: recovery or 90 days, whichever is longer).
- Co-administration: may be given with, before or after any other vaccine, including live vaccines, and with RSV, influenza, pneumococcal or Shingrix on the same day (separate limbs). TST/IGRA may be done any time; retest ≥28 days later if latent TB is strongly suspected.

5. Nuvaxovid vs mRNA: the table for the counter

	Nuvaxovid (XFG)	Spikevax / Comirnaty (2026–27)
What's in it	5 µg ready-made spike protein + Matrix-M saponin adjuvant	mRNA in a lipid nanoparticle; your cells make the spike for a few days
How long in use	Platform decades old (Shingrix, HPV, hep B); COVID version since 2021	Since Dec 2020; billions of doses
Protection	Trials ≈90% vs original strain; comparable to mRNA as a booster	Trials ≈95% vs original strain; ≈50% vs hospitalization early each season
Side effects	Fewer and milder systemic reactions; sore arm common	More fatigue, headache, chills, fever in the first 2 days
Myocarditis	Rare; ≈2 per 100,000 doses reported	Rare; highest in males 12–29 after dose 2 (≈10–20 per 100,000)
PEG allergy	No PEG (contains polysorbate 80)	Contains PEG-2000
Pregnancy	Permitted; less data	Permitted; extensive data
Alberta 2026–27	Funded for the same groups; one dose ≥3 months after last dose	Same

	Nuvaxovid (XFG)	Spikevax / Comirnaty (2026–27)
Storage at the pharmacy	Fridge; prefilled syringes ready to use	Fridge or freezer depending on product; thaw times

Counselling points

- "This is the non-mRNA COVID vaccine. It contains a ready-made piece of the virus's spike protein plus a plant-based ingredient that wakes up the immune system; nothing is made inside your cells and there is no genetic material in it."
- "It's the same kind of vaccine technology as the shingles and hepatitis B shots, and it's been used for COVID since 2021. In the big trials it was about 90% effective, the same as the mRNA vaccines were, and it's updated each year to the current strain, just like they are."
- "Most people get a sore arm; about a third feel tired or achy for a day. Fever and chills are less common than with the mRNA shots. Heart inflammation has been reported very rarely with every COVID vaccine, around 2 in 100,000 with this one, so get checked if you have chest pain, breathlessness or a racing heart in the next few weeks."
- "One dose a year for most people, at least 3 months after your last COVID shot. If you're 80 or older, in a care home, or have a weak immune system, you'll be offered a second dose about 3 months later."
- "If you reacted badly to an mRNA vaccine, or you'd rather not have one, this is the alternative Alberta provides, and you qualify for it free under the same rules."
- "It can be given at the same visit as your flu, RSV, pneumonia or shingles vaccine."

Workflow reminders

- Check prior COVID-19 doses in Netcare; the interval is counted from the last dose of any product. Do not restart a series in immunocompromised patients; count what has been given.
- Prefilled syringe: keep in the carton until use, twist off the tip cap, attach needle (not supplied), give the full 0.5 mL. Vials: swirl, do not shake; label discard time (12 h fridge / 6 h room temperature).
- The monograph still says "two doses 3 weeks apart" for the never-vaccinated and "6 months" after a prior dose; Alberta and NACI use one dose and a 3-month minimum (NACI 2024; AIP 07.229). Follow the AIP.
- Record the product as Nuvaxovid XFG; non-eligible patients pay at the pharmacy (Alberta.ca).

6. Reading the efficacy numbers: what "90% efficacious" means

Vaccine efficacy is a **relative risk reduction** against one pre-defined outcome over a limited follow-up window. It is not "90% of vaccinated people are protected" or "a 10% chance of getting COVID".

The arithmetic, using PREVENT-19

- 29,949 adults randomized 2:1 to Nuvaxovid or placebo; efficacy analysed in those seronegative at baseline (Dunkle et al., NEJM 2022).
- **Endpoint:** first PCR-confirmed COVID-19 with at least one protocol symptom (fever, new cough, shortness of breath, loss of taste/smell, or a cluster of milder symptoms), onset ≥7 days after dose 2; graded mild/moderate/severe.
- 14 cases in ≈17,300 vaccinees vs 63 in ≈8,100 placebo recipients = 3.3 vs 34.0 per 1,000 person-years.
- VE = 1 − (3.3 ÷ 34.0) = **90.4%** (95% CI 82.9–94.6).
- **Absolute terms:** placebo attack rate ≈0.8% over ≈3 months, so the absolute risk reduction was ≈0.7%, roughly one case prevented per 140 people vaccinated during that window at early-2021 US incidence. The relative number travels across settings; the absolute number depends entirely on how much virus is circulating and the patient's baseline risk.
- **What the 90% does not cover:** asymptomatic infection or transmission (swabs were symptom-triggered only); durability beyond ≈3 months; a precise severe-disease estimate (0 vs 4 cases).

Efficacy vs effectiveness. Efficacy comes from a randomized placebo-controlled trial; effectiveness from observational data in routine use (test-negative or cohort designs) and is almost always lower because of waning, variant drift, older and sicker populations, and imperfect case ascertainment. From XBB.1.5 onward every COVID-19 vaccine, Nuvaxovid and mRNA alike, has been authorized on **immunobridging**: a neutralizing-antibody GMT ratio and seroresponse rate versus a previous formulation, with no new efficacy trial (Product monograph).

6.1 What each Nuvaxovid study actually measured

Study	Type	Primary outcome	Result	Not measured / caveat
PREVENT-19 (Dunkle 2022)	Efficacy RCT vs placebo	PCR-confirmed symptomatic COVID-19 ≥7 d after dose 2, baseline seronegative	90.4%; moderate-to-severe 0 vs 10; 92.6% vs variants of concern	Asymptomatic infection; durability >3 mo; severe cases too few
UK phase 3 (Heath 2021)	Efficacy RCT vs placebo	Same definition; severity by FDA criteria	89.7% (80.2–94.6); Alpha 86.3%, non-Alpha 96.4%; severe 0 vs 5	Same; flu co-administration substudy was immunogenicity only
Adolescents 12–17 (Áñez 2023)	Efficacy RCT vs placebo	PCR-confirmed symptomatic COVID-19 ≥7 d after dose 2	79.5% (46.8–92.1), all Delta	20 cases; wide CI
COV-BOOST (Munro 2021)	Phase 2 immunogenicity RCT, MenACWY control	Anti-spike IgG at day 28 as geometric mean ratio vs control; co-primary safety	Nuvaxovid boosted antibody after 2× AstraZeneca and after 2× Pfizer; not in the high-reactogenicity group	**No clinical endpoint**: never counted infections or illness

Study	Type	Primary outcome	Result	Not measured / caveat
Italian cohort (Mateo-Urdiales 2023)	Observational effectiveness	Notified infection; symptomatic COVID-19	31% vs infection; 50% vs symptomatic (Omicron)	Reference was days 3–10 after dose 1; 26 hospitalizations, not estimable
Korean cohort (Gwak 2024, preprint)	Observational vs Pfizer	Any infection; severe (ICU/death)	No consistent difference from mRNA	Hospitalization excluded by policy
XFG and other updated strains	Immuno-bridging	Neutralizing GMT ratio and seroresponse vs prior formulation	Authorized on antibody criteria	No efficacy trial for any 2026–27 COVID-19 vaccine

6.2 The same outcomes for the mRNA vaccines

Caution: these are not head-to-head data. The mRNA trials ran in mid-to-late 2020 against ancestral virus; the Nuvaxovid trials ran in early 2021 against Alpha. Endpoint definitions and timing differ slightly (noted below), so differences of a few percentage points between trials carry no meaning.

	Nuvaxovid	Pfizer-BioNTech (BNT162b2)	Moderna (mRNA-1273)
Trial, size, era	PREVENT-19 n=29,949 and UK n=15,187; Dec 2020–Feb 2021 (Alpha)	C4591001 n=43,548; Jul–Nov 2020, ancestral (Polack, NEJM 2020)	COVE n=30,420; Jul–Oct 2020, ancestral (Baden, NEJM 2021)
Primary endpoint	PCR+ with ≥1 symptom, onset ≥7 d after dose 2, seronegative	NAAT+ with ≥1 symptom, onset ≥7 d after dose 2, no prior infection	PCR+ with ≥2 systemic or ≥1 respiratory symptom, onset ≥14 d after dose 2, seronegative (stricter)
Cases, vaccine vs placebo	14 vs 63 (2:1); UK 10 vs 96	8 vs 162	11 vs 185
Efficacy vs symptomatic COVID-19	90.4% (82.9–94.6); UK 89.7%	95.0% (90.3–97.6)	94.1% (89.3–96.8)
Severe disease	0 vs 4; 0 vs 5 (CI too wide)	1 vs 9 after dose 1; at 6 months 1 vs 30 → 96.7% (80.3–99.9) (Thomas, NEJM 2021)	0 vs 30 → 100%; final analysis 98.2% (El Sahly, NEJM 2021)
Asymptomatic infection	Not measured	Not measured in the primary trial	Swab at dose-2 visit: 14 vs 38 PCR+ without symptoms → ≈63% (post hoc)
Durability in trial	Median ≈3 months	91.3% over 6 months; 96.2% (<2 mo) → 83.7% (4–6 mo)	93.2% over median 5.3 months
Adolescents	12–17: 79.5% (46.8–92.1), Delta	12–15, n=2,260: 0 vs 16 → 100% (75.3–100) (Frenck, NEJM 2021)	12–17, n=3,732: 93.3% (47.9–99.9), 0 vs 4 (Ali, NEJM 2021)
Booster RCT with clinical endpoint	**None**; booster use rests on immunogenicity	Dose 3 vs placebo ≥6 mo after dose 2, n=10,125, Delta: 6 vs 123 → relative efficacy 95.3% (89.5–98.3) (Moreira, NEJM 2022)	None vs placebo; booster authorized on immunogenicity
COV-BOOST (anti-spike IgG GMR vs control, day 28)	Boosted after both primes; moderate reactogenicity	Boosted strongly after both primes	Largest boost of seven vaccines: GMR 32.3 after 2× AZ, 11.5 after 2× Pfizer; higher reactogenicity (Munro 2021)

	Nuvaxovid	Pfizer-BioNTech (BNT162b2)	Moderna (mRNA-1273)
Observational effectiveness, Omicron era	Italy 50% vs symptomatic; hospitalization not estimable; Korea: severe disease no consistent difference vs Pfizer	2024–25 updated vaccines (pooled, mostly mRNA): 51% vs hospitalization at 7–59 d → not measurable at 6–10 mo (Wiegand, JAMA Intern Med 2026); 57% → 34% from 4 to 20 wk (Nebraska cohort)	Same pooled estimates (studies rarely separate the two mRNA products)
2026–27 strain (XFG) basis	Immunobridging	Immunobridging	Immunobridging
Myocarditis reporting rate	≈1.9 per 100,000 doses; pericarditis ≈19 per 100,000 (Clothier 2024)	Males 12–17 after dose 2 ≈7–11 per 100,000; males 18–24 ≈5 per 100,000 (Oster, JAMA 2022)	Males 18–24 after dose 2 ≈5.6 per 100,000 (same source)
Systemic reactogenicity (adult booster)	Mean 1.8 events; 34% with ≥3 (Rousculp 2024)	Mean 3.2 events; 61% with ≥3 (mRNA pooled)	Same pooled estimate; mRNA-1273 generally the more reactogenic

What this tells you

- Against the endpoint everyone quotes (symptomatic PCR-confirmed disease soon after the primary series) all three vaccines landed in the same ≈90–95% band, and Nuvaxovid did so against Alpha rather than the ancestral strain.
- The mRNA vaccines have **more** evidence, not clearly **better** evidence: 6-month trial follow-up, a placebo-controlled booster RCT (Pfizer), an asymptomatic-infection signal (Moderna) and billions of doses of post-marketing data. Nuvaxovid's gaps are trial durability, any booster-efficacy RCT and any hospitalization-effectiveness estimate of its own.
- Severe disease was rare in every trial, so none can separate the products on hospitalization; that comes from observational data, which has largely not separated Nuvaxovid out.
- From XBB.1.5 onward all products are on the same footing: annual updates approved on antibody bridging.
- The evidence reliably separates them only on tolerability: fewer systemic reactions and a lower reported myocarditis rate with Nuvaxovid, which is why NACI treats them as equivalent options rather than ranking mRNA first.

Educational reference for pharmacist practice; confirm eligibility, intervals and supply against the current Alberta Immunization Policy biological page and product monograph.